Supplementary material

Supplementary Table 1 Risk of bias in observational studies.

Reference and publication year	1a	1b	2	3	4	5	6a	6b	7	8	9	10	11	12a	12b	12c	12d	12e	13a	13b	13c	14a	14b	14c	15	16a	16b	16c	17	18	19	20	21	22	Total points
Schoofs [11] 2006	1	1	1	1	1	1	1	NA	1	1	1	0	1	1	1	1	NA	1	1	1	0	1	1	0	1	1	1	NA	1	1	1	1	1	1	28
van Gossum [12] 2009	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	NA	1	1	1	1	1	1	1	1	1	NA	1	1	1	1	1	1	31
Eliakim [13] 2009	1	1	1	1	1	1	1	1	1	1	0	1	1	0	1	1	1	1	1	1	0	1	1	0	1	1	1	?	1	1	1	1	1	0	28
Gay [14] 2010	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	NA	1	1	1	1	1	1	30
Spada [15] 2011	1	1	1	1	1	1	1	NA	1	1	1	1	1	1	1	1	NA	1	1	1	0	1	1	0	1	1	1	NA	1	1	1	1	1	1	29
Herrerìas-Gutiérrez [16] 2011	0	1	1	1	1	0	1	NA	1	1	1	0	0	1	0	0	NA	1	1	1	0	1	0	0	1	?	?	NA	1	1	1	1	1	0	19
Spada [17] 2011	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	1	1	0	0	1	0	1	1	1	0	1	1	1	1	1	1	26
Rondonotti [18] 2014	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	1	1	0	0	1	1	1	1	1	1	29
Hagel [19] 2014	0	1	1	1	1	0	1	1	1	1	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	1	0	0	1	1	0	1	0	0	18
Triantafyllou [20] 2014	1	1	1	1	1	1	1	NA	1	1	1	1	1	1	1	1	1	NA	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1	29
Saito [21] 2015	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	31
Rex [22] 2015	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1	1	0	1	1	1	1	1	1	31
Spada [23] 2015	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	NA	1	1	1	1	1	0	32
Togashi [24] 2015	1	1	1	1	1	1	1	NA	1	1	1	1	1	1	1	1	1	NA	1	0	0	1	1	0	1	1	1	NA	1	1	1	1	1	0	27
Boal Carvalho [25] 2015	0	1	1	1	1	0	1	NA	1	1	0	0	0	1	0	0	0	0	1	0	0	1	0	1	1	1	0	NA	0	1	1	1	1	0	17
Romero-Vazquez [26] 2016	1	1	1	1	1	1	1	NA	1	1	0	0	0	0	0	0	NA	NA	0	1	1	1	0	0	1	1	1	NA	1	1	1	1	1	0	20
Ota [27] 2017	1	1	1	1	1	1	1	N/A	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	?	1	1	1	1	1	0	30
Zhou [28] 2017	1	1	1	1	1	1	1	NA	1	1	0	0	0	0	0	0	NA	NA	1	0	0	0	0	0	1	1	1	NA	0	1	1	1	1	1	18
Nogales [29] 2017	1	1	1	1	1	1	1	NA	1	1	1	0	0	1	0	0	NA	NA	0	0	0	0	0	0	1	1	0	NA	0	1	1	1	1	0	17
Igawa [30] 2017	1	1	1	1	1	1	1	NA	0	0	0	0	0	0	0	0	NA	NA	1	0	0	1	0	0	1	1	1	NA	0	1	1	1	1	0	17
Sato [31] 2017	1	1	1	1	1	1	1	NA	1	1	0	0	1	1	1	0	NA	NA	1	1	0	1	1	0	1	1	1	NA	1	1	1	1	1	1	27
Pioche [32] 2018	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1	1	0	1	1	1	1	1	0	26
Takano [33] 2018	1	1	1	1	1	1	1	NA	1	1	0	1	1	1	1	1	NA	NA	1	1	1	1	1	0	1	1	1	NA	1	1	1	1	1	0	27
Kobaek-Larsen [34] 2018	1	1	1	1	1	1	1	NA	1	1	1	1	1	1	1	1	NA	NA	1	1	1	1	1	0	1	1	1	NA	1	1	1	1	1	0	28
Okabayashi [35] 2018	0	0	1	1	1	1	0	0	1	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	25
Eliakim [36] 2018	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1	1	0	1	1	1	1	1	1	31
Parodi [37] 2018	1	1	1	1	1	1	1	NA	1	1	0	0	1	1	0	0	NA	NA	1	0	0	0	0	0	1	1	0	NA	1	1	1	1	1	0	17
Baltes [38] 2018	1	1	1	1	1	1	1	0	1	1	0	1	1	1	1	0	0	1	1	1	1	1	0	1	1	1	1	0	1	1	1	1	1	1	28
Voska [39] 2019	1	1	1	1	1	1	1	NA	1	1	0	0	0	0	0	0	NA	NA	1	1	0	1	0	0	1	1	1	NA	1	1	1	1	1	0	20
Yamada [40] 2019	0	1	1	1	1	1	0	0	0	0	0	0	0	1	1	0	0	1	1	0	0	1	0	1	1	1	1	0	1	1	1	1	0	1	19
González-Suárez [41] 2020	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	0	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	29
Otani [42] 2020	1	1	1	1	1	1	1	1	1	?	?	1	1	?	1	1	1	1	1	1	1	1	1	?	1	?	1	?	1	1	1	1	1	1	28
Pecere [43] 2020	1	1	1	1	1	1	1	?	1	1	1	1	1	1	1	1	?	1	1	1	?	1	1	1	1	1	1	?	?	1	1	1	1	?	28
Utano [44] 2020	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	?	1	1	1	1	1	1	?	?	1	1	?	1	1	1	1	1	?	29

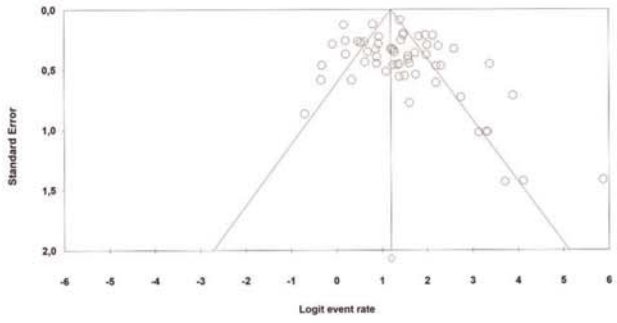
NA, not applicable.

Supplementary material

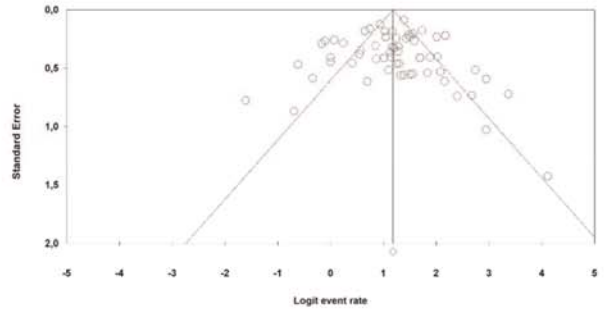
Supplementary Table 2 Risk of bias in randomized controlled trials.

Reference and publication year	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias) (patient-reported outcomes)	Blinding of outcome assessment (detection bias)	Incomplete outcome data addressed (attrition bias) (Short-term outcomes (2-6 weeks)	Incomplete outcome data addressed (attrition bias) (Longer-term outcomes (> 6 weeks)	Selective reporting (reporting bias)	Other bias	No. of low assessments
Eliakim [45] 2006	H	H	H	H	?	L	L	L	?	3
Spada [46] 2011	L	L	L	?	?	L	L	L	L	7
Hartmann [47] 2012	H	H	L	L	?	L	L	L	L	6
Kakugawa [48] 2012	L	H	L	L	NA	L	L	L	?	6
Argüelles-Arias [49] 2014	?	L	?	L	L	L	L	L	?	6
Ramos [50] 2014	L	L	L	L	L	L	L	L	?	8
Brechmann [51] 2016	?	H	L	L	NA	L	L	L	?	5
Kastenberg [52] 2017	L	L	L	L	?	L	L	L	L	7
Alvarez-Urturi [53] 2017	L	L	L	L	NA	L	NA	L	?	6
Buijs [54] 2018	L	L	L	H	NA	L	NA	L	?	5
Kroijer [55] 2018	L	L	L	L	L	L	L	L	?	8
Ohmiya [56] 2018	H	H	H	L	H	L	H	L	?	3

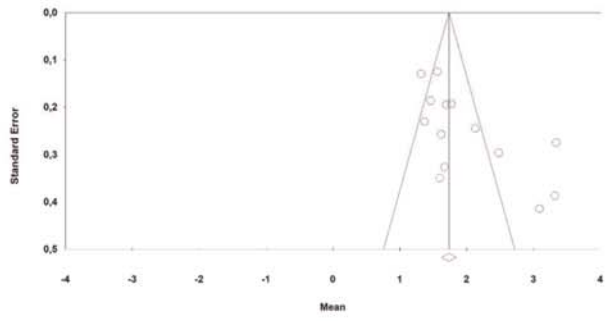
H, high; L, low; NA, not applicable.



Supplementary Figure S1. Funnel plot for completion rates



Supplementary Figure S2. Funnel plot for adequate cleanliness rates



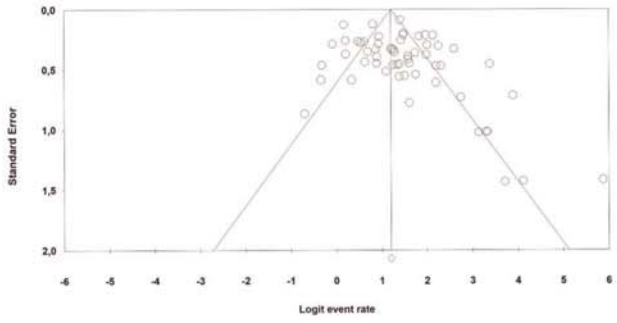
Supplementary Figure S3. Funnel plots for colonic transit times

Supplementary material

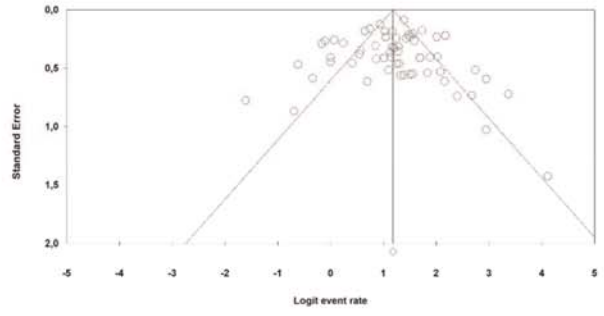
Supplementary Table 2 Risk of bias in randomized controlled trials.

Reference and publication year	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias) (patient-reported outcomes)	Blinding of outcome assessment (detection bias)	Incomplete outcome data addressed (attrition bias) (Short-term outcomes (2-6 weeks)	Incomplete outcome data addressed (attrition bias) (Longer-term outcomes (> 6 weeks)	Selective reporting (reporting bias)	Other bias	No. of low assessments
Eliakim [45] 2006	H	H	H	H	?	L	L	L	?	3
Spada [46] 2011	L	L	L	?	?	L	L	L	L	7
Hartmann [47] 2012	H	H	L	L	?	L	L	L	L	6
Kakugawa [48] 2012	L	H	L	L	NA	L	L	L	?	6
Argüelles-Arias [49] 2014	?	L	?	L	L	L	L	L	?	6
Ramos [50] 2014	L	L	L	L	L	L	L	L	?	8
Brechmann [51] 2016	?	H	L	L	NA	L	L	L	?	5
Kastenberg [52] 2017	L	L	L	L	?	L	L	L	L	7
Alvarez-Urturi [53] 2017	L	L	L	L	NA	L	NA	L	?	6
Buijs [54] 2018	L	L	L	H	NA	L	NA	L	?	5
Kroijer [55] 2018	L	L	L	L	L	L	L	L	?	8
Ohmiya [56] 2018	H	H	H	L	H	L	H	L	?	3

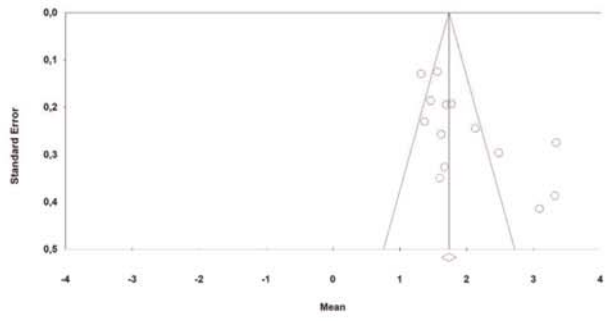
H, high; L, low; NA, not applicable.



Supplementary Figure S1. Funnel plot for completion rates



Supplementary Figure S2. Funnel plot for adequate cleanliness rates



Supplementary Figure S3. Funnel plots for colonic transit times